

Last updated: May 26, 2026

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Education

- Aug. 2021 – **Ph.D. in Artificial Intelligence**, KAIST, Daejeon
Aug. 2025 Advisor: Prof. Jong Chul Ye
- Feb. 2012 – **M.S. in Electrical Engineering**, KAIST, Daejeon
Feb. 2014 Advisor: Prof. Wan Choi
- Mar. 2008 – **B.S. in Media Communication Engineering**, Hanyang University, Seoul
Feb. 2012 *Summa Cum Laude* (GPA: 4.13/4.5)

Work Experience

- Sep. 2025 – **Assistant Professor**, Kookmin University, Seoul
Present Department of Artificial Intelligence
- Feb. 2014 – **Senior Researcher**, Agency for Defense Development (ADD)
Aug. 2021 Wireless communication and network systems for military applications

Research Interests

Generative AI, Trustworthy/Responsible AI, AI for Scientific Discovery,
AI for Defense and Sensing Systems

Publications

* Co-first, † Co-corresponding

- Journal **[J1]** Physics-Guided and Fabrication-Aware Inverse Design of Photonic Devices Using Diffusion Models
Dongjin Seo*, **Soobin Um***, Sangbin Lee, Jong Chul Ye, Haejun Chung
ACS Photonics (2026)
- Conference **[C8]** Beyond Generative Priors: Minority Sampling with JEPA-Guided Diffusion
Sol Park, **Soobin Um**
ICML 2026
- [C7]** Diverse Text-to-Image Generation via Contrastive Noise Optimization
Byungjun Kim*, **Soobin Um***, Jong Chul Ye
ICLR 2026

[C6] DPAIL: Training Diffusion Policy for Adversarial Imitation Learning without Policy Optimization

Yunseon Choi, Minchan Jeong, **Soobin Um**, Kee-Eung Kim
NeurIPS 2025

[C5] Boost-and-Skip: A Simple Guidance-Free Diffusion for Minority Generation

Soobin Um^{*}, Beomsu Kim^{*}, Jong Chul Ye
ICML 2025

[C4] Minority-Focused Text-to-Image Generation via Prompt Optimization

Soobin Um, Jong Chul Ye
CVPR 2025 (Oral presentation, top 0.74%)

[C3] Self-Guided Generation of Minority Samples Using Diffusion Models

Soobin Um, Jong Chul Ye
ECCV 2024

[C2] Don't Play Favorites: Minority Guidance for Diffusion Models

Soobin Um, Suhyeon Lee, Jong Chul Ye
ICLR 2024

[C1] A Fair Generative Model Using LeCam Divergence

Soobin Um, Changho Suh
AAAI 2023 (Oral presentation)

Workshop **[W1]** Physics-guided Optimization of Photonic Structures using Denoising Diffusion Probabilistic Models

Dongjin Seo^{*}, **Soobin Um**^{*}, Sangbin Lee, Jong Chul Ye, Haejun Chung
NeurIPS 2024 Workshop (ML4PS)

Preprint **[P2]** MotionCFG: Boosting Motion Dynamics via Semantic Motion Sharpening

Byungjun Kim, **Soobin Um**[†], Jong Chul Ye[†]
NeurIPS 2026 (under review)

[P1] Self-Improving Diffusion Classifiers with Minority Preference Optimization

Hyunsoo Kim, Jungmyung Wi, **Soobin Um**, Donghyun Kim, Suhyun Kim
ECCV 2026 (under review)

Research Grants & Projects

Apr. 2026 – **Regional Innovation Leading Company Development R&D Program: AI-**
Mar. 2028 **Based Battery State Diagnosis for Electric Two-Wheelers**
(ongoing) **Funded by the MSS (TIPA), Republic of Korea (~KRW 534M / ~USD 400K, Sub-**
contract PI share: ~KRW 100M)
Role: **Subcontract PI**

- Apr. 2026 – **AI S&T Innovation Technology Development Program:** Development of AI for
 Dec. 2029 Multi-Scale Drug Response Transfer Prediction Based on Biological World Models
 (ongoing) Funded by the MSIT (NRF), Republic of Korea (~KRW 3B / ~USD 2.2M, Co-PI
 share: ~KRW 800M)
 Joint research with National Cancer Center
 Role: **Co-Principal Investigator**
- Oct. 2025 – **Future Defense Innovative Technology Program:** AI-Driven Long-Range
 Mar. 2028 EO/IR/SWIR Sensor Fusion System
 (ongoing) Funded by the MSIT, Republic of Korea (~KRW 400M / ~USD 300K)
 Role: **Principal Investigator**
- Apr. 2025 – **AI Star Fellowship Program:** Embodied AI with Multimodal Foundation World
 Dec. 2030 Model for Robotics
 (ongoing) Funded by the MSIT and IITP, Republic of Korea
 Role: Participating Researcher
- Sep. 2024 – **Basic Science Research Program Grant** on Robust Generative AI
 Aug. 2025 Funded by the NRF, Republic of Korea
 Role: **Principal Investigator**
- Jun. 2023 – **Development of AI-Based X-Ray Computer-Based Training Program**
 Jun. 2024 Field-oriented Technology Development Project for Customs Administration
 Funded by the MSIT and Korea Customs Service
 Role: Student Researcher
- Jan. 2023 – **Development of AI Technology for Personalized Plug-and-Play Explanation**
 Jun. 2023 **and Verification of Explanation**
 Funded by the IITP, Korea government (MSIT)
 Role: Student Researcher
- Sep. 2021 – **Development of a Framework to Analyze, Detect, and Mitigate/Remove**
 Aug. 2022 **Bias in AI Models and Training Data**
 Funded by the IITP, Korea government (MSIT)
 Role: Student Researcher

Patents

- Registered **[PR18]** Apparatus, Method, Computer-Readable Storage Medium and Computer
 Program for Assigning Dynamic Frequencies in Wireless Network
 Patent No. 10-2212367, Jan. 2021.
- [PR17]** Method and Apparatus for Satisfaction Degree based Weighted Fair Re-
 source Allocation Optimization in Cognitive Radio Wireless Network
 Patent No. 10-2204935, Jan. 2021.
- [PR16]** Apparatus and Method for Controlling Performance of Receiver for Sub-
 Device in MIMO Cognitive Radio Systems
 Patent No. 10-2192564, Dec. 2020.
- [PR15]** Full Duplex Pair Matching Method for Improving Network Performance in
 Full Duplex Network Environment
 Patent No. 10-2178266, Nov. 2020.

[PR14] Successive-Cancellation Fano Decoding Apparatus and Method for Decoding Using the Same
Patent No. 10-2158312, Sep. 2020.

[PR13] Apparatus and Method for Controlling Channel of Cognitive Radio
Patent No. 10-2107015, Apr. 2020.

[PR12] Data Convergence Method for Reducing Overhead of Cognitive Radio Networks
Patent No. 10-2085205, Feb. 2020.

[PR11] Method and Apparatus for Selecting Frequency Band in Cognitive Radio Network
Patent No. 10-2042260, Nov. 2019.

[PR10] Method and Apparatus for Allocating Frequency Resource in Cognitive Radio Ad-Hoc Network
Patent No. 10-2039650, Oct. 2019.

[PR9] Apparatus and Method for Scheduling Slots for Communication of Data Packets
Patent No. 10-2038051, Oct. 2019.

[PR8] Apparatus and Method for Constructing Rate-compatible Polar Code
Patent No. 10-1996026, Jun. 2019.

[PR7] Time Mirroring Method and System for Airborne Relay Communications
Patent No. 10-1901616, Sep. 2018.

[PR6] Space-Time Dynamic Spectrum Access Apparatus Combined by Multi-Beam Array Antenna and Time Division Duplexing and Frequency Division Duplexing
Patent No. 10-1873102, Jun. 2018.

[PR5] Radio Set System and Setting Channel Method for the Radio Set System
Patent No. 10-1832971, Feb. 2018.

[PR4] Phased Array Antenna System
Patent No. 10-1773481, Aug. 2017.

[PR3] Radio Apparatus for Sensing Space Frequency Spectrum
Patent No. 10-1764655, Jul. 2017.

[PR2] Polarization tracking system using dual polarization antenna with variable gain attenuator and the control method of the same
Patent No. 10-1747789, Jun. 2017.

[PR1] A Dynamic Spectrum Access Technique based on OFDM for P2P Communication
Patent No. 10-1632267, Jun. 2016.

Honors & Awards

May 2026 Silver Reviewer Award, ICML 2026
Aug. 2020 Bronze Medal, National Defense Science Award, ADD

Aug. 2020 Achievement Award, ADD
Jun. 2018 Outstanding Paper Award, KICS Winter Conference
Feb. 2012 *Summa Cum Laude*, Excellence Award, Hanyang University

Invited Talks

Aug. 2025 *MinorityPrompt*, Korean AI Association - KAIA Summer Conference
Aug. 2025 *MinorityPrompt*, Poster Session at KCCV 2025
Jul. 2025 *MinorityPrompt and BnS: Recent Advances in Diffusion-Based Minority Sample Generation*, Visual AI Lab, Princeton University
Jul. 2025 *Physics-Guided and Fabrication-Aware Inverse Design Using Diffusion Models*, The McMahon Lab, Cornell University (Co-presented with Dongjin Seo)
Aug. 2022 *Diffusion Probabilistic Models: A Gentle Introduction*, SpiderCore

Teaching Experience

Instructor

- AI Ethics (Spring 2026, KMU)
- Data Structures (Spring 2026, KMU)
- Generative AI (Fall 2025, KMU)
- Data Science (Fall 2025, KMU)

TA

- AI501: Machine Learning for AI (Spring 2023, KAIST)
- AI619: AI for Medical Imaging (Fall 2022, KAIST, **Head TA**)
- EE424: Introduction to Optimization (Fall 2021, KAIST)
- EE210: Probability and Introductory Random Processes (Fall 2013, KAIST)
- EE321: Communication Engineering (Spring 2013, KAIST)
- Data Science: Modeling for Prediction (Feb. 2022, Hyundai Motors)
- Data Science Master Program (Sep. 2021 – Nov. 2021, Hyundai Motors)

Reviewer Services

Conferences CVPR (2024–2026), ICLR (2025–2026), ICML 2026, ECCV 2026, NeurIPS 2026, WACV 2026
Journals IEEE Transactions on Image Processing (TIP)